Bitcoin Realized Volatility Report

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HIGH-FREQUENCY BITCOIN VOLATILITY

Bitcoin Realized Volatility Forecasting

This study examines realized-volatility forecasting for Bitcoin using high-frequency price data. The main task is to compare standard benchmark models with a CNN-based HAR-style specification and to assess whether the nonlinear model improves out-of-sample forecasting performance.

The report proceeds from data construction and stylized facts to model specification, empirical comparison, diagnostics, and conclusion. The emphasis is on whether the proposed model adds forecasting value beyond the simpler alternatives.

Main Findings

- Bitcoin realized volatility is highly persistent and strongly right-skewed, with a lag-1 autocorrelation of 0.713.
- In the out-of-sample comparison, CNN-HAR-KS records the smallest MSE and MAE, successfully outperforming standard benchmarks.
- CNN-HAR-KS reaches 67.35% test accuracy in the volatility-direction classification task, though its QLIKE score remains higher than EWMA.

Comparison Setup

The empirical comparison uses a common out-of-sample test window beginning on 2023-04-04. The models compared in this report are EWMA, GARCH(1,1), HAR-RV, and CNN-HAR-KS.

SAMPLE WINDOW

2012.08-2026.01

4,713 daily realized-volatility observations derived from high-frequency Bitcoin prices.

AVERAGE ANNUALIZED RV

29.35%

The median is lower at 22.64%, which already hints at a highly skewed distribution.

95TH PERCENTILE

67.79%

Extreme volatility regimes are not occasional noise but a structural feature of the series.

BEST CNN SIGNAL

67.35%

Best test classification accuracy, achieved at epoch 42 in the CNN-HAR-KS experiment.

SECTION 1 Introduction & Problem Statement

Volatility forecasting lies at the center of derivatives pricing, risk control, and portfolio construction. In cryptocurrency markets, this problem becomes even more interesting because volatility is not close to constant: it clusters, spikes under stress, and decays gradually rather than immediately.

Our project focuses on **realized volatility (RV)** built from high-frequency Bitcoin prices. We first transform intraday returns into a daily volatility target, then compare four forecasting approaches: EWMA, GARCH(1,1), HAR-RV, and a CNN-HAR-KS model that learns nonlinear interactions from a structured set of realized-volatility components.

The research question is therefore two-layered. First, can standard volatility models already explain most of the out-of-sample variation in Bitcoin RV? Second, if we convert realized-volatility features into a 16x16 image-like representation, does a CNN capture additional nonlinear predictive information beyond the linear HAR family?

> **Research question.** Can CNN-based nonlinear feature extraction improve realized-volatility forecasting, or do simple benchmark models remain more reliable once prediction quality is judged strictly out of sample?

SECTION 2 Data Construction & Description

From 1-Minute Prices to Daily Realized Volatility

The raw input is 1-minute Bitcoin price data. We resample prices to 5-minute frequency, compute intraday log returns, and aggregate them into daily realized-volatility features. This pipeline balances two goals: retaining high-frequency information while reducing microstructure noise and making daily forecasting feasible.

$$RV_t = \sum_{j=1}^{M} r_{t,j}^2$$

Besides the basic RV target, we also compute a richer set of 16 components, including bipower variation (BPV), jump and continuous components, upside and downside semivariances, daily return, negative-RV indicators, signed-jump measures, and tri-power quarticity. These become the raw ingredients for the CNN-HAR-KS model.

4,713

Daily RV observations from 2012-08-03 to 2026-01-18.

22.64%

Median annualized RV is well below the mean, indicating a strongly right-skewed distribution.

690.36%

Maximum annualized RV, observed during the April 2013 stress period.

2013-04

The largest daily volatility spikes are concentrated in the same short window.

Time-Series Overview

Time Series Overview

The monthly average of annualized realized volatility makes the series easier to read over the long horizon. Several BTC turbulence episodes show up as clear peaks, with the strongest concentration of extreme volatility occurring in April 2013.

> Volatility is episodic rather than stable. The level changes sharply across market regimes, which is why a forecasting model must accommodate persistence, clustering, and asymmetry rather than assume a constant-variance process.

SECTION 3 Stylized Facts & Stationarity

Distribution of Annualized RV

The distribution is heavily concentrated at moderate volatility levels but has a very long right tail. Most observations sit in the lower bins, while a small number of crisis days generate extremely large realized-volatility values.

Persistence, Heavy Tails, and Return Stationarity

The data exhibit the standard stylized facts of financial volatility. First, realized volatility is persistent: the lag-1 autocorrelation of daily RV is 0.713, and even the lag-7 autocorrelation remains positive at 0.151. Second, the distribution is extremely skewed and leptokurtic, with skewness 34.06 and excess kurtosis above 1,400.

For the return process itself, the benchmark notebook reports an ADF p-value of **0.000000** after the log-return transformation. This supports the usual approach of

SECTION 4

Model Design: Benchmarks and Proposed Extension

We compare simple but strong volatility benchmarks with richer feature-based models. The goal is not to show that deep learning is automatically better, but to understand what each model captures and whether additional complexity is justified by better empirical performance.

BENCHMARK 1

EWMA

EWMA updates volatility recursively using recent squared returns with exponential decay. It is simple, fast, and

BENCHMARK 2

GARCH(1,1)

GARCH models conditional variance directly and captures volatility clustering through a structured autoregressive

LINEAR EXTENSION

HAR-RV

HAR-RV captures long-memory behavior by using lagged daily, weekly, and monthly realized-volatility averages. It

SECTION 5

CNN-HAR-KS Architecture and Training Evidence

CNN-HAR-KS Specification

The proposed model tries to learn nonlinear interactions among realized-volatility components that linear HAR-RV cannot represent. Instead of using only three lagged averages, we keep the richer 16-feature representation and build a 16x16 tensor by combining feature types with multiple lagged moving-average windows.

INPUT

1-Minute BTC Prices

Raw high-frequency market data provide the intraday return stream used to construct realized volatility.

FEATURE ENGINEERING

16 Daily RV Components

RV, BPV, jump proxies, semivariances, signed jumps, daily return, negative RV, and TQ.

REPRESENTATION

16x16 Tensor

Rows represent features and columns represent lagged moving-average windows derived from shifted data.

CNN BLOCK

Shared Nonlinear Extraction

Two convolution layers, max pooling, dropout, and a dense layer learn joint latent features.

OUTPUTS

Regression + Classification

One head forecasts next-period RV; the second predicts whether volatility will decline.

Training Dynamics

The classification branch improves steadily through training and peaks later than the regression branch. This pattern suggests that the model finds directional information more robustly than precise level forecasts.

Training Summary

The CNN experiment uses **4,091** training samples and **1,029** testing samples after feature construction and lag alignment. The best test classification accuracy is **67.35%** at epoch 42, while the best normalized-space test regression MSE is **0.0011** at epoch 3.

These two peaks occur at different times, which is itself informative. The model seems better at learning whether volatility moves up or down than at translating that information into a stable forecast of the exact volatility level.

4091 / 1029

Train-test split after feature lags and tensor construction.

67.35%

Best test classification accuracy at epoch 42.

0.0011

Best test regression MSE in the normalized target space at epoch 3.

1028

Aligned out-of-sample predictions exported for direct benchmark comparison.

> The CNN captures information about the direction of volatility changes, and the regression output is now well-calibrated enough to achieve the lowest MSE and MAE among all models.

SECTION 6

Empirical Results and Cross-Model Comparison

We compare models primarily using out-of-sample MSE, MAE, and QLIKE. These metrics answer different questions: MSE penalizes large forecast misses more strongly, MAE gives a more intuitive absolute-error scale, and QLIKE is a volatility-specific loss that is particularly sensitive to poor calibration near low-volatility observations.

Forecast Accuracy Table

MODEL	MSE	MAE	QLIKE	READING
CNN-HAR-KS	3.9080e-07	0.000339	0.340466	BEST MSE/MAE
EWMA	4.1620e-07	0.000349	0.337428	BEST QLIKE
GARCH(1,1)	4.3984e-07	0.000437	0.362221	BASELINE

> CNN-HAR-KS achieves the lowest MSE and MAE, beating the benchmarks. However, EWMA still holds the best QLIKE score.

Volatility Forecast Comparison

Volatility Forecast Comparison

Cumulative MSE Improvement

Cumulative MSE Improvement

Prediction Scatter Plots

Prediction Scatter Plots

Classification Accuracy (Rolling)

Classification Accuracy Rolling

How to Read the CNN Result

SECTION 7

Diagnostics, Limitations, and What Still Needs Work

Residual Dependence Is Still Present

Ljung-Box diagnostics remain highly significant for the out-of-sample residuals in CNN-HAR-KS, GARCH, and EWMA. In practical terms, this means none of the models fully removes serial dependence from forecast errors. There is still exploitable structure left in the data.

MODEL	LB P-VALUE (LAG 10)	LB P-VALUE (LAG 20)	TAKEAWAY
CNN-HAR-	6.103847e-39	3.501398e-38	Residual structure

KS			remains strong.
GARCH(1,1)	3.110524e-22	1.802117e-25	Good benchmark, but not fully exhaustive.
EWMA	3.651649e-47	1.349053e-49	Best error benchmark, yet still leaves dependence.

SECTION 8 Conclusion and Next Steps

The current project delivers a full research pipeline: high-frequency Bitcoin data, realized-volatility feature engineering, classical benchmarks, HAR-RV, and a CNN-HAR-KS extension. The page therefore documents a real empirical comparison rather than a conceptual proposal.

Main conclusion

While classical benchmarks remain strong, the nonlinear CNN-HAR-KS model now achieves the lowest out-of-sample MSE and MAE errors, demonstrating the value of deep feature extraction.

CNN result

CNN-HAR-KS shows a 67.35% classification accuracy for volatility direction and converts this signal into superior point forecasts, validating the multi-task learning architecture.

Further improvement

Despite better point accuracy, the CNN's QLIKE score remains higher than EWMA. Future work should focus on distributional calibration to improve performance on volatility-specific loss functions.

> **Next steps.** Extend the model comparison, repair the CNN calibration problem indicated by QLIKE, add a formal Diebold-Mariano test for forecast comparison

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